Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 04)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

Food .....

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Fashion .....

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Entertainment / pets .....

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Furniture and ornaments .....

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Traditional medicines .....

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0 1 . 2 Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

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2 .....

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3 .....

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0 1 . 3 Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

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0 2 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

0 2 . 1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer ____ %

0 2 . 2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

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0 2 . 3 Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

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0 2 . 4 Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

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Extra space

15

0 3 . 1 Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

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0 3 . 2 Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

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0 3 . 3 Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

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0 3 . 4 Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

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0 4 Table 1 shows the estimated population of blue whales and the number killed by whaling in selected years.

Table 1

Year	1900	1930	1960	1986	2020
Estimated population	240 000	100 000	5 000	2 000	15 000
Whales killed that year	200	30 000	1 200	0	0

0 4 . 1 Calculate the percentage decline in the blue whale population between 1900 and 1986.

[1 mark]

Answer ☐ %

0 4 . 2 Explain why whale populations were so easily over-exploited.

[3 marks]

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0 4 . 3 Describe how the International Whaling Commission regulates whaling.

[4 marks]

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0 4 . 4 Explain why the ITTO has been less successful than the IWC in achieving sustainable exploitation.

[2 marks]

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0 5 . 1 State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

0 5 . 2 Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

0 5 . 3 Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

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0 5 . 4 Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

10

0 6 . 1 Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

A Tropical rainforest ☐

B Oceanic islands ☐

C Temperate broadleaf woodland ☐

D Mangroves ☐

0 6 . 2 Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

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0 7 . 1 Explain the effects of deforestation on soil.

[4 marks]

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0 7 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

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0 7 . 3 Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

1 .....

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2 .....

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0 8 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 8 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 8 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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0 9 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 9 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 9 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

1 .....

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1 0 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

1 0 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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1 0 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

1 0 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

1 0 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

11 . 1 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

11 . 2 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS